

Abnormal fixational eye movements in strabismus

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ABSTRACT

Introduction Fixational saccades are miniature eye movements that constantly change the gaze during attempted visual fixation. Visually guided saccades and fixational saccades represent an oculomotor continuum and are produced by common neural machinery. Patients with strabismus have disconjugate binocular horizontal saccades. We examined the stability and variability of eye position during fixation in patients with strabismus and correlated the severity of fixational instability with strabismus angle and binocular vision.

Methods Eye movements were measured in 13 patients with strabismus and 16 controls during fixation and visually guided saccades under monocular viewing conditions. Fixational saccades and intersaccadic drifts were analysed in the viewing and non-viewing eye of patients with strabismus and controls.

Results We found an increase in fixational instability in patients with strabismus compared with controls. We also found an increase in the disconjugacy of fixational saccades and intrasaccadic ocular drift in patients with strabismus compared with controls. The disconjugacy was worse in patients with large-angle strabismus and absent stereopsis. There was an increase in eye position variance during drifts in patients with strabismus. Our findings suggest that both fixational saccades and intersaccadic drifts are abnormal and likely contribute to the fixational instability in patients with strabismus.

Discussion Fixational instability could be a useful tool for mass screenings of children to diagnose strabismus in the absence of amblyopia and latent nystagmus. The increased disconjugacy of fixational eye movements and visually guided saccades in patients with strabismus reflects the disruption of the fine-tuning of the motor and visual systems responsible for achieving binocular fusion in these patients.

INTRODUCTION

The ability to perceive depth is a tightly coupled process between the visual afferent and efferent systems. Normally, the two eyes see slightly different images (binocular disparity); however, the stimulation of the respective retina of each eye produces a visual sensation of a common subjective visual direction (fusion) relative to the fovea in a phenomenon called correspondence. Retinal images with similar size and shape can be fused. Stereopsis occurs in the presence of binocular disparity that is not large enough to cause diplopia. The achievement of fusion and stereopsis is complex because during attempted visual fixation, our eyes are never completely still. Physiological involuntary fixational eye movements comprise microsaccades, drifts and ocular tremor.^{1–3} Despite the constant motion of the eyes, the normal fixational eye movements do

not prevent binocular fusion suggestive of close coordination between the visual sensory and motor systems. Multiple studies have reported greater fixational instability in amblyopia.^{4–7} A few studies have reported increased fixation instability in patients with strabismus.^{8–10} Ciufredda *et al*^{8,9} showed that latent nystagmus and saccadic intrusions were seen in patients with strabismus, whereas increased drift was related to the presence of amblyopia. Their studies had a very small cohort of patients with strabismus alone. Increased positional variability of the fixing and non-fixing eye has been reported in patients with exotropia.¹⁰ Monkeys with strabismus also have an increase in the fixation scatter of the fixing and non-fixing eye.^{11,12} This increased fixational instability may be related to altered microsaccade alone, drift alone or both microsaccades and drifts. None of the studies to date have characterised fixational instability in patients with strabismus without latent nystagmus. Thus, the first purpose of our study was to quantitatively characterise the abnormalities in fixational saccades and intersaccadic drifts in subjects with varying severity of strabismus.

Patients with strabismus can have disconjugate horizontal saccades.^{13–15} Disconjugate saccades have also been reported in monkeys with strabismus.¹⁶ Saccades and microsaccades follow an oculomotor continuum,¹⁷ and both types of movements are generated by common neural circuits.^{18–21} Besides saccades, strabismus is known to produce asymmetric, variable and subnormal convergence response.²² Hence, we anticipate that fixational saccades will be disconjugate in patients with strabismus, which can further disrupt the achievement of the binocular vision. We hypothesised that patients with large-angle strabismus and absent stereopsis would have greater disconjugacy of both fixational eye movements and visually guided saccades.

METHODS

High-resolution video-based eye tracker (EyeLink 1000, SR Research, Ontario, Canada) was used to non-invasively measure horizontal and vertical eye position in subjects with strabismus and controls. The experiment protocol complied with the tenets of the Declaration of Helsinki and was approved by the Cleveland Clinic's Institutional Review Board. All subjects provided a written informed consent. The written consent was obtained from the parents or legal guardians on behalf of the minors or children enrolled in the study.

Subject's head was supported on a chin rest, 55 cm away from the liquid-crystal display screen. The subjects were instructed to fixate their gaze on a red circular visual target (the diameter subtended 0.5°



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Table 1 Clinical and demographic features

Subject	Age	Acuity	Acuity Left Eye	Stereopsis	Refraction OD	Refraction OS	Strabismus Primary near	Strabismus Primary distance
S1	14	0	0.09	Absent	Plano	Plano	16 residual XT, s/p strab surgery	16 residual XT, s/p strab surgery
S2	9	0	0	Absent	+3.25	+3.25	40 ET	35 ET
S3	13	0	0	Present	-3.0	-3.0	25 X (T)	30 X (T)
S4	58	0	0	Absent	-1.0	-0.75	50 XT	50 XT
S5	47	0	0	Absent	-4.5	-1.25	14 consecutive XT-s/p strab surgery	14 consecutive XT-s/p strabismus surgery
S6	9	-0.12	-0.12	Present	+0.50	+0.50	8-10 X (T)	8-10 X (T)
S7	9	0.09	0.09	Absent	+1.25	+1.25	16 residual XT, s/p strab surgery	16 residual XT, s/p strabismus surgery
S8	19	0	0	Absent	-0.25	Plano	50 XT	50 XT
S9	54	0	0	Absent	-0.25	Plano	50 residual XT-s/p strabismus surgery	50 residual XT-s/p strabismus surgery
S10	17	0	0	Absent	Plano	-1.75	40 X (T)	45 X (T)
S11	11	0	0	Present	-1.25	-0.50	10-12 X (T)	10-12 X (T)
S12	61	0	0	Present	-3.25	-3.50	16 ET	16 ET
S13	6	0	0	Present	+0.5	+0.50	6-8 X (T)	8-10 X (T)
N1	36	0	0	Present	-2.5	-2.5	Ortho	Ortho
N2	31	0	0	Present	-3.5	-3.5	Ortho	Ortho
N3	26	0	0	Present	-4.0	-4.5	Ortho	Ortho
N4	7	0	0	Present	-2.5	-2.5	Ortho	Ortho
N5	24	0	0	Present	-3.5	-3.25	Ortho	Ortho
N6	24	0	0	Present	-2.5	-2.5	Ortho	Ortho
N7	42	0	0	Present	-1.5	-0.75	Ortho	Ortho
N8	8	0	0	Present	+0.5	+0.75	Ortho	Ortho
N9	9	0	0	Present	+0.25	+0.50	Ortho	Ortho
N10	27	0	0	Present	-8.0	-7p.5	Ortho	Ortho
N11	24	0	0	Present	-2.5	-2.5	Ortho	Ortho
N12	34	0	0	Present	-6.5	-7.25	Ortho	Ortho
N13	13	0	0	Present	+0.75	+0.75	Ortho	Ortho
N14	31	0	0	Present	-2.5	-2.5	Ortho	Ortho
N15	62	0	0	Present	-1.5	-1.75	Ortho	Ortho
N16	64	0	0	Present	-2.25	-2.25	Ortho	Ortho

ET, constant esotropia; N, Normal control; S, Strabismic subject; s/p= status post strabismus surgery; XT, constant exotropia.

visual angle) projected on the screen with a white background in a completely dark room for 45 s. We measured fixations at primary (0°) position and horizontal saccades of $\pm 5^\circ$, $\pm 10^\circ$ and $\pm 15^\circ$ amplitudes. We recorded binocular eye positions under monocular viewing conditions by using an infrared filter, which blocks the visible light over one eye. Each eye was calibrated under monocular viewing conditions before the beginning of a trial. Subjects wore corrective lenses for the experiments.

We recruited 13 subjects with strabismus (eight girls, five boys) and sixteen controls (eight girls, eight boys) (table 1). None of the patients with strabismus had latent nystagmus or amblyopia. Stereopsis was assessed using the Titmus Fly Stereotest. Subjects with strabismus were grouped based on their strabismus angle (small angle: <15 prism dioptres (PD), $n=4$; medium angle: $15-30$ PD, $n=4$; and large angle: >30 PD, $n=5$) and stereopsis status (stereopsis present: if they could at least detect the fly on the Stereotest or were better, $n=5$; and stereopsis absent: if they were not able to detect the fly on the Stereotest, $n=8$).

Eye position data were used for further analysis. Blinks and partial blinks were identified and removed.^{23 24} Fixational saccades defined as saccades produced during attempted fixation were identified using the unsupervised clustering method²⁵ (see online Supplementary file 1 and supplementary materials).

Saccade amplitude was defined as the absolute difference between the eye positions at the start and at the end of the saccade. Drifts were defined as epochs between fixational saccades and blinks.

The composite vector amplitude of horizontal and vertical displacements during fixational saccades, eye velocity and eye position variability during intersaccadic drifts was computed for the viewing and non-viewing eye. Saccade amplitude and drift velocity disconjugacies were calculated as a difference in these parameters between the viewing and non-viewing eye.

We quantified the fixation stability by measuring a bivariate contour ellipse area (BCEA) using the following equation.²⁶

$$\text{BCEA} = \pi X^2 \sigma_x \sigma_y \sqrt{1 - p^2}$$

where 2.291 is the X^2 value (2 df) corresponding to a probability of 0.68, σ_x and σ_y are the horizontal and vertical SD of eye position, respectively, and p is the product moment correlation of two position components.

The data analysis was performed using custom-prepared scripts in Matlab and GraphPad Prism six software for statistical analysis. A one-way analysis of variance (ANOVA) test was used to compare the disconjugacy of fixational saccades,

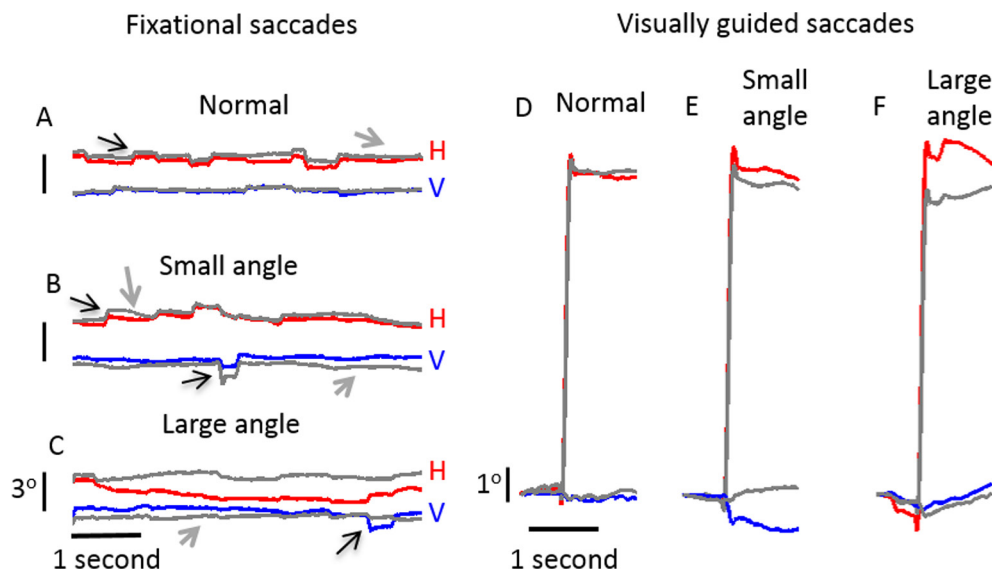


Figure 1 An example of visual fixation during a 5 s epoch under conditions of monocular viewing from a normal control (A), small-angle strabismus and stereopsis present (B) and large-angle strabismus and absent stereopsis (C). An example of a rightward visually guided saccade under conditions of monocular viewing from a normal control (D), small-angle strabismus and stereopsis present (E) and large-angle strabismus and absent stereopsis (F). The x-axis represents time, and the y-axis represents horizontal (red: right eye, grey: left eye) and vertical (blue: right eye, grey: left eye) eye positions. Black arrows represent fixational saccades, whereas grey arrows represent intrasaccadic drift.

visually guided saccades and intersaccadic drift velocity across subjects. A Kruskal-Wallis ANOVA test was used to compare the frequency and amplitude of fixational saccade, eye position variance of drifts and composite drift velocity across subjects.

RESULTS

Patients with strabismus have fixation disparity with misalignment between the right and left eye and abnormal binocular function. Therefore, we investigated the dynamics of fixational saccades, visually guided saccades and intersaccadic drifts in patients with strabismus as a function of severity of strabismus and presence of stereopsis. We assessed whether the fixational saccades and ocular drifts are disconjugate in strabismus and compared the disconjugacy of fixational saccades with visually guided saccades.

Figure 1 depicts visual fixation in a control subject (N8) (figure 1A), in a subject with small-angle strabismus and stereopsis present (S13) (figure 1B) and a subject with large-angle strabismus and no stereopsis (S10) (figure 1C) under monocular viewing conditions. During a 5 s epoch, the control subject had comparable amplitudes between the right and left eye during both fixational saccade and intersaccadic drift (figure 1A). The subject with small-angle strabismus (figure 1B) had a mild disconjugacy during both fixational saccade and intersaccadic drift whereas the subject with large-angle strabismus and no stereopsis had greater disconjugacy during both fixational saccade and intersaccadic drift (figure 1C). We found a similar increase in disconjugacy in a patient with large-angle strabismus during a visually guided saccade (figure 1D–F). We further quantified these results in the subsequent sections.

Figure 2A summarises the comparison of composite amplitude disconjugacy of the fixational saccades in controls and those with strabismus. There was an increase in the disconjugacy as a function of the strabismus angle (controls: $0.17^\circ \pm 0.24^\circ$; small-angle strabismus: $0.22^\circ \pm 0.29^\circ$; medium-angle strabismus: $0.52^\circ \pm 1.41^\circ$ and large-angle strabismus: $0.54^\circ \pm 1.4^\circ$,

one-way ANOVA $p < 0.0001$). A post hoc Dunnett's multiple comparisons tests showed that the medium-angle and large-angle subjects with strabismus differed significantly at $p < 0.05$ from the controls. A similar result was seen when comparing horizontal and vertical fixational saccade amplitude disconjugacy among the four groups (see online Supplementary table 1).

Figure 2B summarises the comparison of disconjugacy of the visually guided saccades in controls and subjects with strabismus. Similar to the fixational saccades, the disconjugacy of horizontal visually guided saccades increased with an increase in the strabismus angle (controls: $0.50^\circ \pm 0.50^\circ$; small-angle strabismus: $0.52^\circ \pm 0.53^\circ$; medium-angle strabismus: $1.3^\circ \pm 1.8^\circ$; large-angle strabismus: $2.79^\circ \pm 3.4^\circ$, one-way ANOVA $p < 0.0001$). The fixational saccade and visually guided saccade disconjugacy also correlated with the size of fixational saccade (Pearson correlation $r = 0.14$, $p < 0.001$) and visually guided saccade (Pearson correlation $r = 0.2$, $p = 0.002$) amplitude, respectively, with disconjugacy being more pronounced for larger amplitude saccades.

Figure 2C summarises the comparison of composite amplitude disconjugacy of the fixational saccades in controls, strabismic subjects with and without stereopsis (controls: $0.17^\circ \pm 0.24^\circ$; strabism with stereopsis: $0.49^\circ \pm 1.34^\circ$; strabism without stereopsis: $0.43^\circ \pm 1.27^\circ$, one-way ANOVA $p < 0.0001$). A post hoc Dunnett's multiple comparisons tests showed that both strabism with and without stereopsis differed significantly at $p < 0.05$ from the controls.

Figure 2D summarises the comparison of disconjugacy of the visually guided saccades in controls and strabismic subjects with and without stereopsis. The amount of disconjugacy of the visually guided saccades was greater in strabismic subjects with absent stereopsis (controls: $0.50^\circ \pm 0.50^\circ$; strabism with stereopsis: $0.95^\circ \pm 1.81^\circ$; strabism with absent stereopsis: $1.98^\circ \pm 2.82^\circ$, one-way ANOVA $p < 0.0001$).

We computed the frequency and number of square wave jerks (SWJs), composite amplitude and directional trajectory of fixational saccades in the viewing and non-viewing eye as a

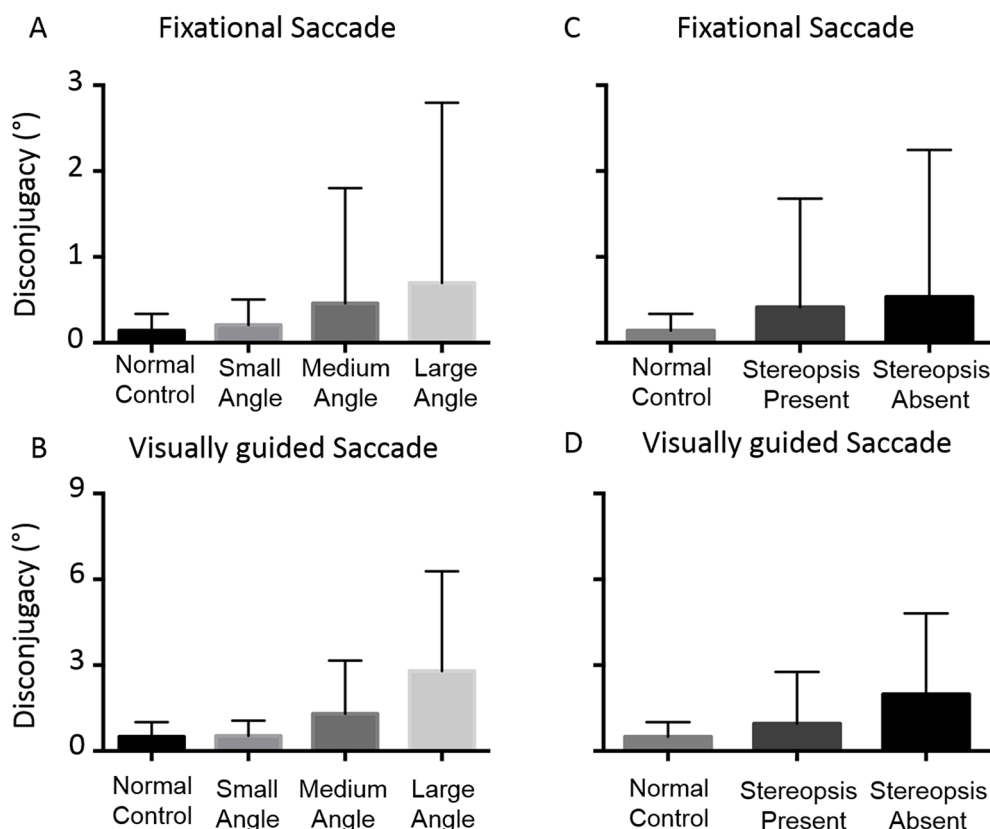


Figure 2 Bar graphs plotting mean and SD of amplitude disconjugacy of fixational saccade (A) and visually guided saccade (B) as a function of strabismus angle. Bar graphs plotting mean and SD of amplitude disconjugacy of fixational saccade (C) and visually guided saccade (D) as a function of presence or absence of stereopsis.

function of strabismus angle and stereopsis status (table 2). The frequencies and the number of SWJ were comparable in controls and subjects with strabismus in the viewing and non-viewing eye. Subjects with strabismus had greater amplitude in the non-viewing eye compared with controls.

We assessed the kinematics of fixational and visually guided saccades by comparing their displacement amplitude with the peak velocity in the main sequence analysis. Figure 3A and B plots the main sequence of the saccades in the viewing and

non-viewing eye, respectively, as a function of strabismus angle, whereas figure 3C and D plots the main sequence as a function of stereopsis. The fixational saccades follow the same main sequence relationship as the visually guided saccades in all subjects with strabismus irrespective of the strabismus angle or presence of stereopsis. Also, the relationship is comparable to that of controls. The normal main sequence in patients with strabismus also suggests that the strabismus is not caused by cranial nerve palsies.

Table 2 Characteristics of fixational saccades in subjects with strabismus as a function of strabismus angle and stereopsis status

Fixational saccades	Controls	Small angle	Medium angle	Large angle	ANOVA, p value
Strabismus angle					
Frequency (Hz)	1.17±0.44	0.83±0.18	1.17±0.19	1.08±0.64	0.30 (KW)
No of SWJs	25±14	13±8	22±8	15±11	0.14 (KW)
Amplitude (°) VE	0.60±0.54	0.90 ± 0.79*	0.90 ± 0.71*	0.60±0.75	<0.0001
Amplitude (°) NVE	0.58±0.53	0.93 ± 0.76*	1.30 ± 1.53*	0.95 ± 1.69*	<0.0001
Direction (°) VE	182±92	169±95	184±96	180±96	0.17
Direction (°) NVE	180±91	166±93	184±101	178±96	0.07
Stereopsis status					
Frequency (Hz) VE	1.17±0.44	1.09±0.23		1.05±0.52	0.42 (KW)
No of SWJs	25±14	21±7		15±10	0.07 (KW)
Amplitude (°) VE	0.60±0.54	0.98 ± 0.76*		0.65±0.75	<0.0001
Amplitude (°) NVE	0.58±0.53	1.31 ± 1.47*		0.93 ± 1.48*	<0.0001
Direction (°) VE	182±92	183±97		176±95	0.9
Direction (°) NVE	180±91	179±101		175±95	0.7

*These groups were significantly different at $p < 0.05$ compared with controls on post hoc Dunnett's multiple comparisons test.

ANOVA, analysis of variance; NVE, non-viewing eye; VE, viewing eye; SWJs, square wave jerks.

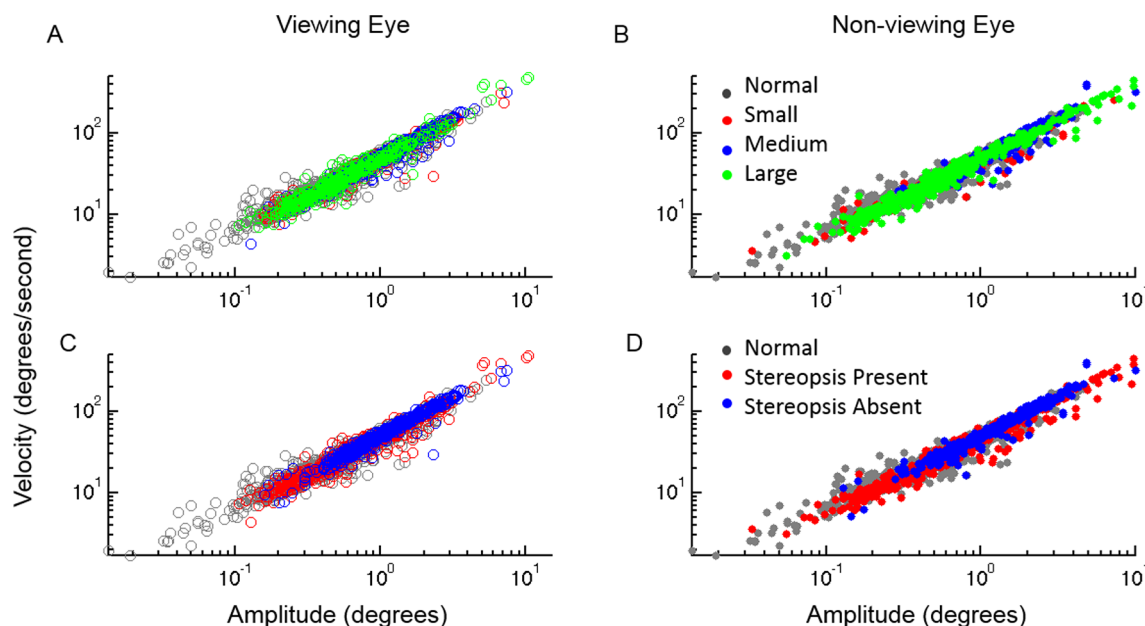


Figure 3 Comparison of amplitude to velocity relationship of fixational saccades in the viewing and non-viewing eye of patients with strabismus and normal control. (A) and (B) plots this relationship as a function of strabismus angle whereas (C) and (D) plots this relationship as a function of binocularity status. All the symbols in patients with strabismus and normal controls fall on the same continuum.

We analysed the inter-saccadic movement in the gaze (drifts) during visual fixation. We first assessed the disconjugacy in the drift velocity; [figure 4A](#) summarises the comparison of controls and subjects with strabismus. The drift velocity disconjugacy increased with an increase in strabismus angle (controls: $0.76^\circ/\text{s} \pm 0.87^\circ/\text{s}$; small-angle strabismus: $0.83^\circ/\text{s} \pm 1.3^\circ/\text{s}$; medium-angle strabismus: $1.09^\circ/\text{s} \pm 2.51^\circ/\text{s}$; large-angle strabismus: $1.17^\circ/\text{s} \pm 1.2^\circ/\text{s}$, one-way ANOVA $p < 0.0001$). [Figure 4B](#) summarises the comparison of disconjugacy in the drift velocity in controls and strabismic subjects with and without stereopsis. The disconjugacy in drift velocity was higher in subjects with strabismus compared with controls. There was no difference in the disconjugacy of drift velocity between patients with or without stereopsis (controls: $0.76^\circ/\text{s} \pm 0.87^\circ/\text{s}$; patients with stereopsis: $1.09^\circ/\text{s} \pm 2.38^\circ/\text{s}$; patients without stereopsis: $1.03^\circ/\text{s} \pm 1.29^\circ/\text{s}$, one-way ANOVA $p < 0.0001$).

We analysed the drift variance in the viewing and non-viewing eye of subjects with strabismus and compared them with controls ([table 3](#)). Subjects with strabismus had higher drift variance compared with controls. The drift variance was greater in the non-viewing eye compared with the viewing eye in subjects with strabismus. Patients with large-angle strabismus as well as absent stereopsis had greater drift variance. Similarly, we measured the variance of pupil size in both normal (0.07 ± 0.05) and subjects with strabismus (0.08 ± 0.06) and found no difference between the two groups (unpaired t-test, $p = 0.1$).

We measured the BCEA of the viewing and non-viewing eye as a function of strabismus angle and stereopsis status ([table 4](#)). The best fixation stability was evident in controls, the fixation instability of the non-viewing eye worsened with increasing angle of strabismus and in patients with absent stereopsis.

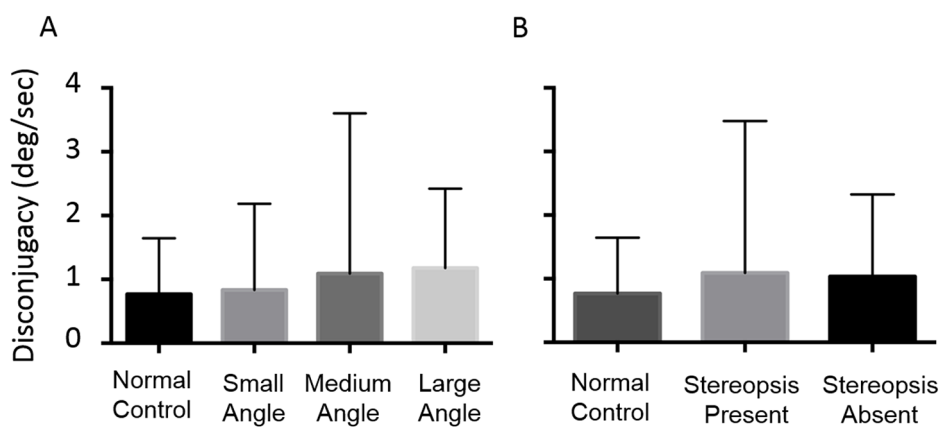


Figure 4 Bar graphs plotting mean and SD of disconjugacy of intersaccadic drift velocity as a function of strabismus angle (A). Bar graphs plotting mean and SD of disconjugacy of intersaccadic drift velocity as a function of presence or absence of stereopsis (B).

Table 3 Characteristics of ocular drifts in subjects with strabismus as a function of strabismus angle and stereopsis status

Composite eye position variance (°)	Controls	Small angle	Medium angle	Large angle	Kruskal-Wallis ANOVA, p value
Strabismus angle					
Viewing eye	0.009±0.04	0.01±0.10	0.01±0.02	0.01±0.04	<0.0001
Non-viewing eye	0.009±0.039	0.02±0.07	0.07±0.17	0.06±0.35	<0.0001
Stereopsis status					
Viewing eye	0.009±0.04	0.014±0.08		0.014±0.03	0.05
Non-viewing eye	0.009±0.039	0.03±0.15		0.06±0.28	<0.001

ANOVA, analysis of variance.

Table 4 Bivariate contour ellipse analysis as a function of strabismus angle and stereopsis status

	Controls	Small angle	Medium angle	Large angle	Kruskal-Wallis ANOVA, p value
Strabismus angle					
Viewing eye	0.30	0.64	0.71	0.90	0.17
Non-viewing eye	0.28	0.87	1.73	3.24	0.0005
Stereopsis status					
Viewing eye	0.30	0.63		0.84	0.07
Non-viewing eye	0.28	1.362		2.478	<0.0002

ANOVA, analysis of variance.

DISCUSSION

The main findings of our study are: (1) patients with strabismus had greater fixation instability in the deviating eye; (2) patients with strabismus had higher intersaccadic drift velocity in the viewing and non-viewing eye; (3) subjects with strabismus had greater disconjugacy in the fixational saccades, visually guided saccades and intrasaccadic eye drifts; and (4) the fixational and visually guided saccades in patients with strabismus had equivalent amplitude and velocity relationship similar to normal.

Increased fixation instability has been reported in patients with strabismus as well as in monkeys with strabismus.^{8–10,12} Fixational stability can be affected due to the presence of latent nystagmus in strabismus.^{27,28} We report the presence of fixational instability in patients with strabismus who do not have the classic ocular motor constellation of infantile strabismus that comprises latent nystagmus, nasotemporal pursuit asymmetry and dissociated vertical deviations. Our study reports increased fixation scatter in both the viewing and non-viewing eye (although greater in the non-viewing eye) of patients with strabismus compared with controls. The increased fixation scatter is due to a combination of an increase in the amplitude of fixational saccades and an increase in eye position variance of the intersaccadic drift. None of the studies to date have characterised fixational instability as a function of strabismus angle. We found that patients with small-angle strabismus and preserved binocular function had better fixational stability whereas patients with large-angle strabismus and absent stereopsis had the highest fixation scatter. Thus, the data collectively suggested that the presence of strabismus alone is sufficient to disrupt the fixational stability even in the absence of amblyopia and latent nystagmus.

Previous studies in monkeys with strabismus ascribed the fixational instability to abnormal development of vergence eye movements.^{22,29} The stimulation of the rostral pole of the superior colliculus changes the strabismic angle in monkeys, putatively via affecting the vergence neurons. Recent physiological and computational studies have suggested that the balance of reciprocal inhibition between omnipause and long-lead burst neurons modulated by the projections from the superior colliculus may play a role in triggering microsaccadic or saccadic

events.^{18–20} Thus, the increase in the fixation scatter could be ascribed to the role of the superior colliculus in the generation of the fixational saccades as well as the presence of vergence neurons within it that are responsible for maintaining the state of eye misalignment.

Several studies have shown that the same circuit generates microsaccades as well as visually guided saccades.^{30,31} Previous studies have reported increased disconjugacy of visually guided saccades in patients with strabismus.¹⁴ Disconjugate saccades have been reported following electrical stimulation of the superior colliculus in monkeys with strabismus.^{32–34} Thus, these studies have concluded that disruption in saccadic machinery either within the superior colliculus or in the downstream areas are responsible for driving the saccadic disconjugacy. Consistent with this notion, we found comparable disconjugacy in fixational and visually guided saccades in patients with strabismus. We also found that fixational saccades in patients with strabismus and controls have comparable amplitude and velocities, suggesting equally efficacious burst generation of fixational saccades in the presence of strabismus. Thus, our results suggest that the fixational saccades and visually guided saccades retain the ocular motor continuum in strabismic states similar to that reported in normals.^{17,31,35–38}

The steady-state misalignment in patients with strabismus could be due to both the disconjugate saccades and disconjugate drifts. The degree of disconjugacy depends on the strabismus angle and binocular vision. Thus, it is possible that a series of ocular motor abnormalities in patients with strabismus disrupt the fine-tuning of the motor and visual systems responsible for achieving binocular fusion.

Contributors Conceptualization: FG and AGS. Data collection and analysis: FG and AGS. Funding acquisition: FG. Writing, reviewing and editing: FG, JO-M and AGS.

Competing interests None declared.

Patient consent Obtained.

Ethics approval Cleveland Clinic IRB.

Provenance and peer review Not commissioned; externally peer reviewed.

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